METRIC

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SUPERSEDING
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PERFORMANCE SPECIFICATION

DYE, LIQUID FOR THE DETECTION OF LEAKS IN AIRCRAFT FUEL SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for liquid dyes which are added to aircraft fuel systems to aid in the detection of fuel leaks (see 6.1).

1.2 Classification. Liquid dye will be of the following types as specified (see 6.2.)

Type II – Yellow

Type III – Yellow/Green Fluorescent

Comments, suggestions, or questions on this document should be addressed to AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB, OH 45433 or e-mailed to AFPET.PTPS@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6820

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	–	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-83133	–	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

29 CFR 1910.1200	–	Hazard Communication

(Copies of this document are available online at https://ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	–	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from https://asq.org.)

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ASTM INTERNATIONAL

ASTM D56	–	Standard Test Method for Flash Point by Tag Closed Cup Tester
ASTM D92	–	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D97	–	Standard Test Method for Pour Point of Petroleum Products
ASTM D1310	–	Standard Test Methods for Flash Point and Fire Point of Liquids by Tag Open-Cup Apparatus
ASTM D1655	–	Standard Specification for Aviation Turbine Fuels
ASTM D2276	–	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM D3241	–	Standard Test Method for Thermal Oxidation Stability of Aviation Turbine Fuels
ASTM D3278	–	Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus
ASTM D4057	–	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4177	–	Standard Practice for Automatic Sampling of Petroleum and Petroleum Products
ASTM D5452	–	Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration
ASTM D5949	–	Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure pulsing Method)
ASTM D6304	–	Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration
ASTM E275	–	Standard Practice for Describing and Measuring Performance of Ultraviolet and Visible Spectrophotometers
ASTM E1064	–	Standard Test Method for Water in Organic Liquids by Coulometric Karl Fischer Titration

(Copies of these documents are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Materials. The dyes shall consist essentially of liquid oil soluble dyes, completely organic and containing no reactive amino groups. The contractor shall certify that no carcinogenic or potentially carcinogenic constituents are present as defined under the Hazard Communications Standard (HCS) (29 CFR 1910.1200) and shall be made available to the contracting officer or the contracting officer's designated representative. The composition of the dye shall be made available upon request of the using activity. The product shall be entirely suitable for the intended use (see 6.1).

3.3 Chemical and physical requirements. The chemical and physical requirements of the finished dyes shall conform to the requirements listed in section 3, Table I, and Table II when tested in accordance with the applicable test methods. Requirements contained herein are not subject to correction for test tolerances.

3.4 Storage stability. The finished dyes shall remain stable and conform to the requirements in section 3, Table I, and Table II, as applicable, for at least one year when tested in accordance with 4.4.5. This test is applicable only for the first article sample.

3.5 Thermal stability. In addition to the requirements of Table I and Table II, Type II (Yellow) and Type III (Yellow/Green Fluorescent) dyes shall not degrade the thermal stability of MIL-DTL-5624, grade JP-5, or MIL-DTL-83133, grade JP-8, fuels, when tested in accordance with ASTM D3241 (see 4.4.2). Should JP-8 not be available, NATO F-24 (ASTM D1655 Grade Jet A additized per the additization requirements for JP-8 per MIL-DTL-83133) can be used instead as an acceptable alternative.

3.6 Color. The liquid dye shall be Type II (Yellow) or Type III (Yellow/Green Fluorescent) as specified (see 6.2).

3.7 Workmanship. When examined visually, the liquid dye shall be completely homogeneous material free from any insoluble matter.

TABLE I. Physical requirements and test methods for Type II Yellow Dye.

Requirements	Minimum	Maximum	Test Method
Moisture, mass percent		0.08	D6304 or E1064
Flash point, open cup, °C	32		D92 or D1310
Flash point, closed cup, °C	25		D56 or D3278
Pour point, °C		-18	D97 or D5949
Color value (absorptivity)	24		1/
Wavelength at maximum absorption, nanometers (nm)	423	442	1/
Contamination of aircraft fuel, mg/L		1.0	D2276 or D5452 2/
Notes: 1/ See 4.4.3 for test procedure. 2/ See 4.4.1 for test procedure.			

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TABLE II. Physical requirements and test methods for Type III Yellow/Green Fluorescent Dye.

Requirements	Minimum	Maximum	Test Method
Moisture, mass percent		0.08	D6304 or E1064
Flash point, open cup, °C	43		D92 or D1310
Flash point, closed cup, °C	40		D56 or D3278
Pour point, °C		-18	D97 or D5949
Percent transmittance at 525 nanometers (nm)			
a. Dilution ratio dye: Xylene = 1:1250		80	1/
b. Dilution ratio dye: Xylene = 1:833		71	1/
Contamination of aircraft fuel, mg/L		1.0	D2276 or D5452 2/
Notes: 1/ See 4.4.4 for test procedure. 2/ See 4.4.1 for test procedure.			

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. All the test requirements of section 3, including 3.5 (thermal stability), shall be completed prior to bid on any product required under this specification.

4.2.1 First article sample**.** When a contractor has not previously furnished material under this specification, one liter (1 L) of the liquid dye from the first batch that represents the product offered to the Government under any contract shall be forwarded to the Aerospace Fuels Laboratory, AFPET/PTPLA, 2430 C Street, Building 70 Area B, Wright Patterson AFB OH 45433. Test data which show the results of tests required by section 3 shall accompany the sample submitted for first article testing.

4.3 Conformance inspection. The conformance inspection (Government Lot Acceptance Testing) shall be conducted on samples from randomly selected containers of each packaged lot (see 6.3.2) and consists of tests to verify conformance to all the requirements in section 3, to include Table I and Table II, except storage stability. Samples shall be labeled completely with information identifying the purposes of the sample, name of the product, specification number, lot and batch number, date of manufacture, date of sampling, and contract number.

4.3.1 Sampling.

4.3.1.1 Sampling for the examination of filled containers. Take a random sample of filled containers from each lot in accordance with ASQ/ANSI Z1.4.

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4.3.1.2 Sampling for tests. Take samples from bulk or packaged lots for tests in accordance with ASTM D4057 or ASTM D4177, or both, except where individual test procedures contain specific sampling instructions.

4.4 Inspection test methods. Tests to determine conformance to chemical and physical requirements shall be conducted in accordance with ASTM Standards, using the applicable methods as listed in Table I, Table II, and 4.4.1 through 4.4.5.

4.4.1 Particulate contamination. Aviation turbine fuel conforming to MIL-DTL-5624, grade JP-5, shall be filtered through a Millipore 0.45 micrometer membrane, or equivalent. Dye the prefiltered fuel in the ratio of 0.57 gram of dye to 3.78 L (1 gallon) of fuel, assuring a uniform mixture. The contamination resulting from the dissolved liquid dye shall be determined in accordance with ASTM D2276 or ASTM D5452. Repeat this test using prefiltered fuel conforming to MIL-DTL-83133, grade JP-8. Should JP-8 not be available, NATO F-24 can be used instead as an acceptable alternative.

4.4.2 Thermal stability. Aviation turbine fuel conforming to MIL-DTL-5624, grade JP-5, or MIL-DTL-83133, grade JP-8, shall be blended in ratios of 0.45 gram of dye to 3.78 L (1 gallon) of fuel. When testing in accordance with ASTM D3241, the thermal stability of the dyed JP-5 or JP-8 fuels shall meet MIL-DTL-5624 or MIL-DTL-83133 requirements, as applicable. Should JP-8 not be available, NATO F-24 can be used instead as an acceptable alternative.

4.4.3 Color value and wavelength of maximum absorption. Accurately weigh to 0.0001 gram a 0.4200 gram to 0.4400 gram sample of yellow dye and transfer to a 100 mL volumetric flask. Any amount within the given ranges may be taken. Dilute to the mark with spectrometric grade xylene). Agitate until the solution is uniform. Using a class A pipette, transfer 2.00 mL of this solution into a 250 mL volumetric flask and dilute to a mark with iso-octane. Record the visible spectrum of the final solution in a 1.000 cm cell versus that of iso-octane employing a spectrophotometer which has been checked for wavelength accuracy and photometric accuracy by ASTM E275. Determine the wavelength of maximum absorbance by inspection of the curve. Calculate the absorptivity of a liquid dye using expression:

$$a = \frac{A}{(bc)}$$

Where:

a = absorptivity

A = absorbance of the solution at the wavelength of maximum absorbance

b = cell thickness in centimeters (cm)

c = concentration of the liquid dye in iso-octane in grams per liter. If the recommended procedure is followed, then the concentration of the measure dye solution in grams per liter is equal to the sample weight in grams divided by 12.5 liters.

4.4.4 Percent transmittance measurement. Accurately weigh 0.1392 gram to the nearest 0.0001 gram of fluorescent dye in a 200 mL volumetric flask and dilute to the mark with spectrometric grade xylene to give a 1250 X dilution. Agitate until the solution is uniform. Fill a 2.54 cm diameter glass tube cell with this solution. Record the percent transmittance of this solution using a spectrophotometer at 525 nm (nanometer).

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Repeat the above procedure but this time accurately weigh 0.2089 gram to the nearest 0.0001 gram of fluorescent dye in a 200 mL volumetric flask and dilute to the mark with spectrometric grade xylene to give a 833 X dilution. Agitate until the solution is uniform.

4.4.5 Storage stability. Store a one-liter sample of the liquid dye in its original container at room temperature of 25 °C ± 3 °C for 12 months. After 12 months, shake the can vigorously by hand and examine visually for evidence of product deterioration or can corrosion. After the visual examination, subject the stored sample to all examinations and tests of this specification.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The liquid dyes covered by this specification are intended to aid in the detection of aircraft fuel system leaks by coloring aircraft fuels. Fuel system leaks accrue in military aircraft due to the extreme stress conditions the airframes encountered. Commercial aircraft are not subject to these stress conditions. There is no currently available non-Government standard which fulfills the requirements of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and color of dye required (see 3.6).
c. Packaging requirements (see 5.1).
d. Quantity required and size containers desired.
e. One copy of the certificate of analysis that lists those items in Table I and Table II, as applicable, should be forwarded to the acquisition activity.

6.3 Definitions.

6.3.1 Bulk lot. A bulk lot consists of an indefinite quantity of a homogeneous mixture of liquid dye offered for acceptance in a single, isolated container; or manufactured in a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in the ingredient materials.

6.3.2 Packaged lot. A packaged lot consists of an indefinite number of one-gallon containers or smaller unit packages of identical size and type, offered for acceptance, and that have been filled with a homogeneous mixture of liquid dye from one isolated container or filled with a homogenous mixture of material manufactured in a single plant run (not to exceed 24 hours), through the same processing equipment, with no change in the ingredient materials.

6.4 Safety. Adequate ventilation should be provided to remove the highly flammable and toxic solvent vapors.

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6.5 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 Subject term (key word) listing.

Aviation Turbine fuel
JP-5
JP-8
JP-8 + 100
NATO F-24

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

a. Main changes in this revision:

i. Removal of Type I (red). Type I is removed because it is no longer authorized as leak detection dye due to operational and safety problems associated with its use.

ii. Addition of NATO F-24 as an acceptable alternative to JP-8 should JP-8 not be available as test fuel.

iii. Addition of additional test methods in Table I and Table II.

iv. Revised sections 3.4 (Storage stability), 3.5 (Thermal stability), 4.3 (Conformance inspection), 4.4.3 (Color value and wavelength of maximum absorption), and 4.4.4 (Percent transmittance measurement).

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CONCLUDING MATERIAL:

Custodians:
- Army – AV
- Navy – AS
- Air Force – 68
- DLA – GS

Review activities:
- Army – AT, EA, MR, SM
- Navy – CG, SH
- Air Force – 11, 19
- DLA – PS

Preparing activity:
- Air Force - 68
- (Project 6820-2023-001)

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